

Victor Le - DA 401: Does Human-Rated Chess Puzzle Difficulty Predict Large Language Model Performance?

Research Design - Revised

Research Design and Study Objective

This study will examine whether chess puzzles that are difficult for humans are also difficult for large language models and which characteristics of a chess position help explain differences between human and LLM difficulty. Human difficulty will be represented by Lichess puzzle ratings, while LLM performance will primarily be measured by whether each model successfully solves a puzzle. The study will then examine whether characteristics such as tactical theme, solution length, number of legal moves, game phase, and number of pieces help explain LLM performance beyond the human difficulty rating. These variables follow the research gap identified in my literature review, where previous studies have generally focused either on predicting human puzzle difficulty or evaluating overall LLM chess ability rather than directly examining where the two disagree.

The study will be observational rather than causal. Although I will control how puzzles are presented to the LLMs, characteristics such as puzzle rating, tactical theme, and solution length already exist in the data and are not randomly assigned. Therefore, the results will show which features are associated with LLM success or failure, but they will not establish that a particular feature directly causes a model to fail.

Dataset

The main dataset will come from the Lichess open puzzle database. The study will use approximately 1,500-2,000 puzzles from the Lichess open puzzle database. The final sample size will depend on the cost and runtime observed during the pilot study, but the same sampled puzzles will be evaluated by every LLM.

Each Lichess puzzle contains information such as the board position, solution moves, puzzle rating, tactical themes, and other metadata. Puzzle rating will serve as the main measure of human difficulty because the rating is based on the performance of many human solvers rather than the judgment of a small number of experts. Previous research has similarly used Lichess or ChessTempo solving data to measure human puzzle difficulty (Ding et al., 2024; Hristova et al., 2014).

I will first remove incomplete or unusable records and reconstruct each puzzle position using Python and the python-chess library. I will then create a stratified sample across the puzzle-rating distribution so that the dataset contains sufficient numbers of easier, medium, and harder puzzles based on Lichess puzzle rating. I will also examine the representation of common tactical themes and different solution lengths before beginning LLM testing.

Data and Large Language Model Preparation

The main independent variable will be Lichess puzzle rating, which will be treated as a continuous measure of human difficulty. Rating groups may also be created for descriptive graphs so that LLM performance can be compared across easier and harder sections of the rating distribution.

Several additional variables will describe the characteristics of each chess position. These will include tactical theme, solution length, number of legal moves, tactical themes, number of pieces remaining, material balance, and game phase.

Other variables will be calculated directly from the chess position using python-chess. For example, the number of legal moves will measure how many possible choices are available to the solver, while piece count and material balance will provide basic measures of board complexity. These choices are also supported by previous work on chess difficulty. Björkqvist (2025), for example, considered variables including legal moves, solution length, tactical themes, material conditions, mobility, sacrifices, and mating threats when predicting human puzzle difficulty.

The main dependent variable will be whether an LLM solves the puzzle correctly. A correct solution will be coded as 1 and an incorrect solution as 0. I will also keep more detailed information about failures, including whether the model produces an illegal move, a legal but incorrect move, or an answer that cannot be interpreted. This distinction may be useful because previous LLM chess research has identified problems with board understanding, move legality, tactical calculation, and producing the correct final move even after plausible reasoning (Wen et al., 2025; Liu et al., 2026).

Three LLMs will be evaluated in this study: GPT-5.6 Sol, Qwen 3 8B, Llama 3.1 8B. Each puzzle will begin in a new conversation through a Python loop so that previous puzzles cannot be accessed and affect later responses.

All models will receive the same basic prompt and the same representation of the chess position whenever possible. The exact board representation will be finalized during a pilot test of approximately 50-100 puzzles. This pilot will also be used to confirm the response format, evaluate cost and runtime, and identify problems with parsing model outputs before the full experiment begins. This follows the pilot stage already included in my original project schedule. The representation of the board is important because previous work has shown that LLM chess performance can change depending on whether legal moves are provided, how the board is represented, and how the model is instructed (Kolasani et al., 2025).

For each puzzle, the model will be asked to provide its next move in a standardized format. For multi-step puzzles, a correct model move will be followed by the opponent's response from the official puzzle solution, after which the model will be asked for its next move. The process will continue until the puzzle is completed or the model makes an incorrect move. A puzzle will be considered fully solved only if the model correctly completes the required solution sequence. This allows multi-step puzzles to remain in the study rather than restricting the sample to single move problems.

Statistical and Machine Learning Analysis

The analysis will begin with descriptive statistics showing how each LLM performs across different types of puzzles. I will calculate overall solving accuracy and compare accuracy across puzzle ratings, tactical themes, solution lengths, game phases, and other position characteristics. These comparisons will first show whether LLM performance generally decreases as human-rated difficulty increases.

The main statistical method will be logistic regression because the primary outcome is whether an LLM solves a puzzle correctly or incorrectly. Because each puzzle will be attempted by multiple LLMs, the responses from all models will be pooled into a single dataset, with one observation for each puzzle and model combination. Model identity will be included as a categorical variable to account for differences in overall performance between LLMs. Since observations from the same puzzle are related, standard errors will be clustered by puzzle. I will first examine the relationship between Lichess puzzle rating and LLM success. I will then add other position characteristics such as solution length, number of legal moves, tactical themes, game phase, and piece count, ... This will show whether these variables provide additional information about LLM performance after human difficulty has already been taken into account.

$$\log \left(\frac{p_{ij}}{1 - p_{ij}} \right) = \beta_0 + \beta_1 \text{LichessRating}_i + \beta_2 \text{SolutionLength}_i + \beta_3 \text{LegalMoves}_i \\ + \beta_4 \text{PieceCount}_i + \beta_5 \text{MaterialBalance}_i + \beta_6 \text{TacticalTheme}_i + \beta_7 \text{GamePhase}_i + \gamma_j$$

p_{ij} = probability that LLM j correctly solves puzzle i

i = puzzle index

j = LLM index

β_0 = intercept

$\beta_1, \dots, \beta_7$ = coefficients for the puzzle characteristics

γ_j = effect of LLM identity

As a secondary machine learning method, I will use XGBoost classification to predict whether an LLM will solve each puzzle. XGBoost is a tree-based machine learning model that builds a sequence of decision trees, with later trees improving on errors made by earlier ones (Chen & Guestrin, 2016). It is useful for this project because it can capture more complicated patterns and interactions between puzzle characteristics. For example, solution length may have a stronger relationship with LLM performance for some tactical themes than for others.

The logistic regression and XGBoost models will serve different purposes. Logistic regression will make it easier to interpret how individual variables are associated with LLM success, while XGBoost will provide a more flexible predictive model and help identify which features are most useful for prediction. The data will be divided into training and testing sets at the puzzle level, so all observations associated with the same puzzle remain in the same partition, and I will compare model performance using measures such as classification accuracy and ROC-AUC. I will also examine XGBoost feature importance to determine which puzzle characteristics contribute most to its predictions.

References

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